

COMPE 510 – PA 4 Report

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Results

Part 1: Data Preprocessing

Fprintf result of part 1

```
===== Part 1: Data Preprocessing =====  
Loading data ...  
First 10 examples from the training dataset:  
x = [1 96 64 27 87 33 0 21], y = 0  
x = [13 126 90 0 0 43 1 42], y = 1  
x = [1 117 88 24 145 34 0 40], y = 1  
x = [5 111 72 28 0 24 0 27], y = 0  
x = [0 101 62 0 0 22 0 25], y = 0  
x = [1 135 54 0 0 27 1 62], y = 0  
x = [1 144 82 40 0 41 1 28], y = 0  
x = [1 180 0 0 0 43 0 41], y = 1  
x = [6 99 60 19 54 27 0 32], y = 0  
x = [2 146 76 35 194 38 0 29], y = 0  
  
Normalizing Features ...  
First 10 examples from the training dataset after normalization:  
x = [-1 -1 -0 0 0 0 -1 -1], y = 0  
x = [3 0 1 -1 -1 1 0 1], y = 1  
x = [-1 -0 1 0 1 0 -0 1], y = 1  
x = [0 -0 0 0 -1 -1 -0 -1], y = 0  
x = [-1 -1 -0 -1 -1 -1 -0 -1], y = 0  
x = [-1 0 -1 -1 -1 -1 1 2], y = 0  
x = [-1 1 1 1 -1 1 0 -0], y = 0  
x = [-1 2 -3 -1 -1 1 -1 1], y = 1  
x = [1 -1 -0 -0 -0 -1 0 -0], y = 0  
x = [-1 1 0 1 1 1 -0 -0], y = 0  
  
Program paused. Press enter to continue.
```

Part 2: Train Neural Network

Fprintf result of part 2

```
===== Part 2: Train Neural Network =====
betal computed from gradient descent:
-2.665967
-6.905695
-8.539135
-9.928722
-7.876420
-9.506057
-7.646958
-7.784057
-2.650980
-2.145439
-11.796039
-0.891585
-11.087877
-5.461849
-14.764354
-11.675144
-5.069424
-5.267752
-3.762284
-7.083220
-4.096176
6.418517
-8.250503
-2.373858
3.717279
6.471248
-7.301801
-2.307978
0.154515
-5.075827
-4.962639
3.264908
7.757028
-4.355091
2.108020
-3.946992
-11.429098
9.745718
0.907664
5.330127
-2.517669

beta2 computed from gradient descent:
7.734200
7.625823
13.702173
17.539384
14.805276
-14.865899
14.267907
-12.075935
-8.624329
-15.846648
6.110829
15.398828
-18.938423
14.606347
-17.352972
-13.387183
-11.797942
10.956361
11.935043
-10.620282
-16.488748

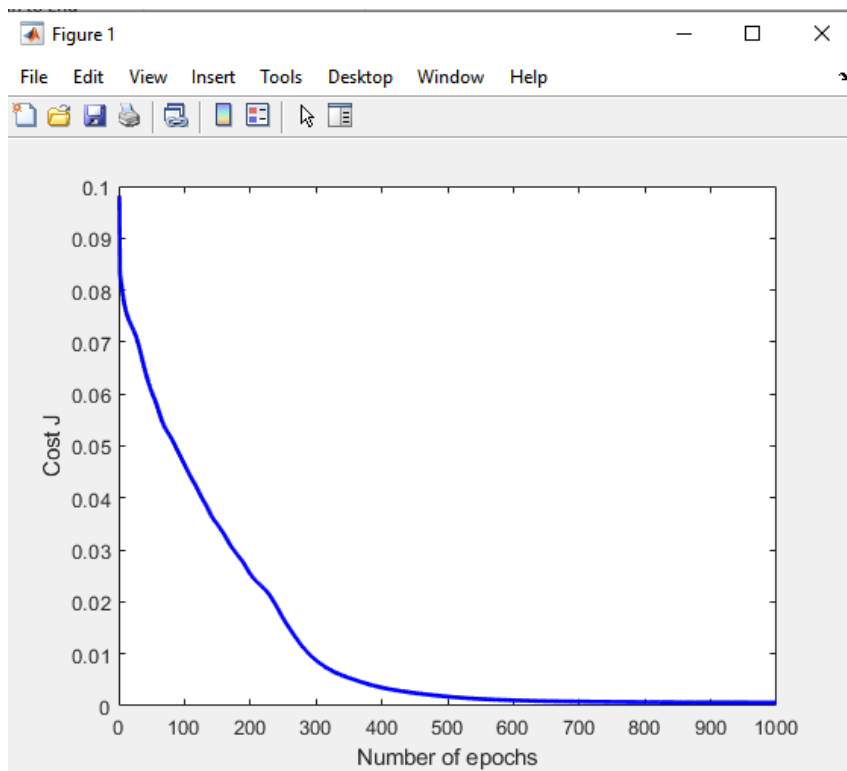
Program paused. Press enter to continue.
```

Part 3: Evaluate Performance

Fprintf result of part 3

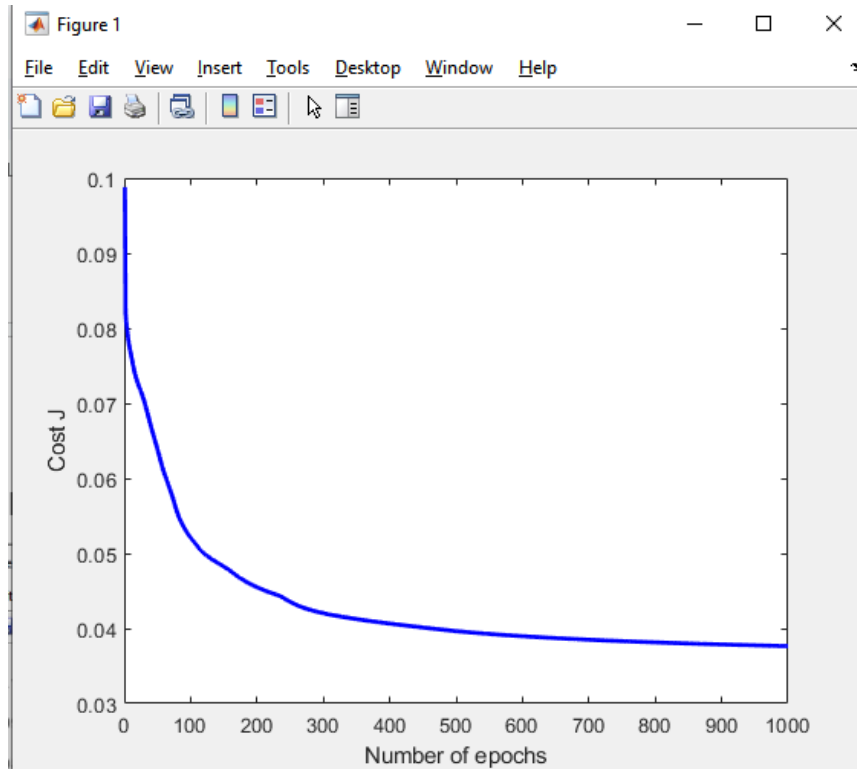
```
==== Part 3: Evaluate Performance ====  
Accuracy:  
65.800866  
  
Predicted output:  
0
```

Graph with $\alpha = 0.1$, num_epochs = 1000, num_hidden = 20



```
Accuracy:  
65.800866
```

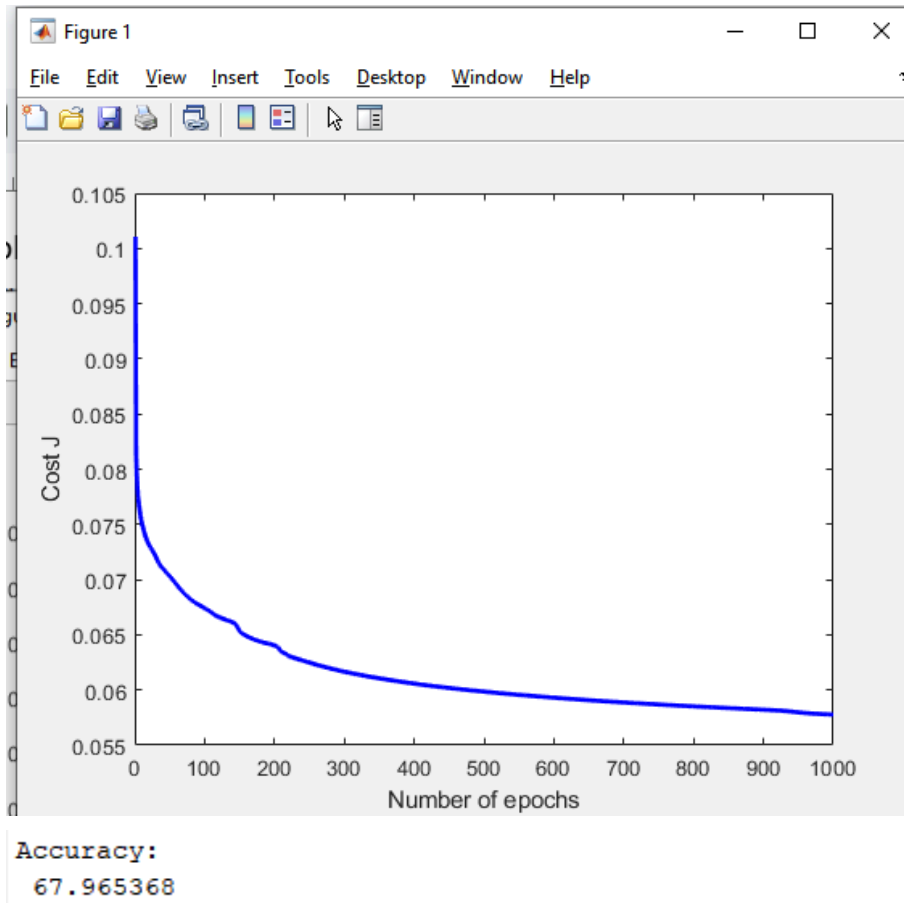
Graph with $\alpha = 0.1$, $\text{num_epochs} = 1000$, $\text{num_hidden} = 10$



Accuracy:

70.129870

Graph with $\alpha = 0.1$, $\text{num_epochs} = 1000$, $\text{num_hidden} = 5$



Questions:

1. The number of hidden units affects the model's performance negatively, with more hidden units decreasing the accuracy (overfitting), but too few hidden units also results in less accuracy likely due to not having enough capacity to capture the patterns in the data. There seems to be a sweet spot with 10 hidden units, giving the best accuracy of about 70%.
2. The model generalizes well, with the accuracy staying around 70%, the model successfully learned patterns from the training data and the predictions were accurate on the testing set.
3. The previous logistic regression model was slightly better than the neural network model – it was noticeably quicker to run and produced slightly higher accuracy (around 75% compared to 70%). The logistic regression model is also significantly less complex, and has a smoother convergence graph compared to the neural network model.

Implementation

Implementing the neural network went well, backpropagation is a useful technique for neural networks given training data. While the neural network didn't completely outperform the logistic regression model, the principles will be useful for data sets and outputs that logistic regression is unable to handle.